AQA 3.1.8 Thermodynamics: lattice enthalpies

AQA A-level Chemistry · Revision questions · Sheet 3 of 3

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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23 marks · about 25 minutes

Key information

- Larger lattice enthalpy: higher ionic charges and smaller ions give stronger attraction
- Perfect ionic model: ions are perfect spheres / point charges with no covalent character
- Born–Haber value bigger than perfect ionic value → covalent character (small, highly charged cation polarises the anion)
- Lattice formation: $M^+(g) + X^-(g) \rightarrow MX(s)$, exothermic; lattice dissociation is the reverse and endothermic
- First IE: $X(g) \rightarrow X^+(g) + e^-$; first EA: $X(g) + e^- \rightarrow X^-(g)$ (exothermic)

Section A: Short-answer questions

01 Explain why the enthalpy of lattice formation of magnesium oxide is much more exothermic than that of sodium fluoride.

[2 marks]

02 The experimental (Born–Haber) value for the enthalpy of lattice formation of magnesium iodide is more exothermic than the value calculated using a perfect ionic model. Explain why.

[2 marks]

03 State and explain which of sodium chloride and potassium chloride has the more exothermic enthalpy of lattice formation.

[2 marks]

04 Explain why all enthalpies of lattice dissociation are endothermic.

[1 mark]

05 Which equation represents the enthalpy of lattice formation of calcium fluoride?

[1 mark]

A $Ca(s) + F_2(g) \rightarrow CaF_2(s)$

B $Ca^{2+}(g) + 2F^-(g) \rightarrow CaF_2(s)$

C $Ca^{2+}(aq) + 2F^-(aq) \rightarrow CaF_2(s)$

D $CaF_2(s) \rightarrow Ca^{2+}(g) + 2F^-(g)$

06 Which compound has the most exothermic enthalpy of lattice formation?

[1 mark]

A NaCl

B KBr

C CaO

D MgO

07 Suggest why enthalpies of lattice formation are found using Born–Haber cycles rather than measured directly.

[1 mark]

08 Give an equation, including state symbols, for the process that represents the second ionisation energy of magnesium.

[1 mark]

Section B: Longer questions

09 This question is about the enthalpy changes in a Born–Haber cycle for sodium chloride.

09.1 Give an equation, including state symbols, for the enthalpy of atomisation of sodium.

[1 mark]

09.2 Give an equation, including state symbols, for the first ionisation energy of sodium.

[1 mark]

09.3 Give an equation, including state symbols, for the enthalpy of formation of sodium chloride.

[1 mark]

09.4 Of the enthalpy of atomisation of sodium, the first ionisation energy of sodium, the first electron affinity of chlorine and the enthalpy of lattice formation of sodium chloride, state which **two** are exothermic.

[2 marks]

09.5 Explain why the first ionisation energy of sodium is endothermic.

[1 mark]

10 The table shows lattice enthalpies based on a perfect ionic model and lattice enthalpies from Born–Haber cycles for three metal chlorides.

Lattice enthalpy of dissociation / kJ mol^{-1}	Perfect ionic model	Born–Haber cycle
Calcium chloride	2223	2237
Potassium chloride	690	701
Silver chloride	770	905

Discuss the values in the table. In your answer you should

- compare the three values based on a perfect ionic model
- compare the values based on a perfect ionic model to the values from a Born–Haber cycle for each compound.

[6 marks]

Mark Scheme

M1, M2... are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	M1 Mg^{2+} and O^{2-} have higher charges (than Na^+ and F^-) M2 stronger attraction between the ions	*Source: Practice question (written for this set)*	2
02	M1 Mg^{2+} is small / has a high charge density and polarises / distorts the (large) I^- ion M2 so the bonding has some covalent character, which strengthens the lattice	*Source: Practice question (written for this set)*	2
03	M1 sodium chloride M2 Na^+ is smaller than K^+ so stronger attraction to Cl^-	*Source: Practice question (written for this set)*	2
04	Energy is needed to overcome the attraction between oppositely charged ions	*Source: Practice question (written for this set)*	1
05	**B**	*Source: Practice question (written for this set)*	1
06	**D**	*Source: Practice question (written for this set)*	1
07	Gaseous ions cannot be combined directly / the process cannot be carried out (measured) experimentally	*Source: Practice question (written for this set)*	1
08	$Mg^+(g) \rightarrow Mg^{2+}(g) + e^-$	*Source: Practice question (written for this set)*	1
09.1	$Na(s) \rightarrow Na(g)$	*Source: Practice question (written for this set)*	1
09.2	$Na(g) \rightarrow Na^+(g) + e^-$		1
09.3	$Na(s) + \frac{1}{2}Cl_2(g) \rightarrow NaCl(s)$		1
09.4	first electron affinity of chlorine enthalpy of lattice formation (of NaCl)		2
09.5	Energy is needed to overcome the attraction between the (outer) electron and the nucleus		1
10	**Level 3 (5–6 marks)** All stages covered and generally correct and virtually complete; coherent, logical progression from stage 1 to 2 and 3. **Level 2 (3–4 marks)** All stages covered but incomplete / with inaccuracies, OR two stages generally correct and virtually complete. **Level 1 (1–2 marks)** Two stages covered but incomplete / inaccurate, OR one stage generally correct and virtually complete. **Level 0** Insufficient correct chemistry.	*Source: 7405/1 Jun 2023 Q05.7* **Stage 1** 1a $CaCl_2$ value larger / KCl and AgCl similar; 1b Ca^{2+} has a larger charge / smaller ion; 1c $CaCl_2$ has stronger attraction between + and – ions **Stage 2** 2a $CaCl_2$ values similar; 2b KCl values similar; 2c $CaCl_2$ and KCl (almost) perfectly ionic / ions are point charges **Stage 3** 3a AgCl larger difference; 3b AgCl has (some) covalent character; 3c Ag^+ more polarising / distorts the electron cloud more	6